Exam Date & Time: 25-Sep-2024 (10:45 AM - 12:15 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

B.Tech III Semester Sessional Examination September 2024

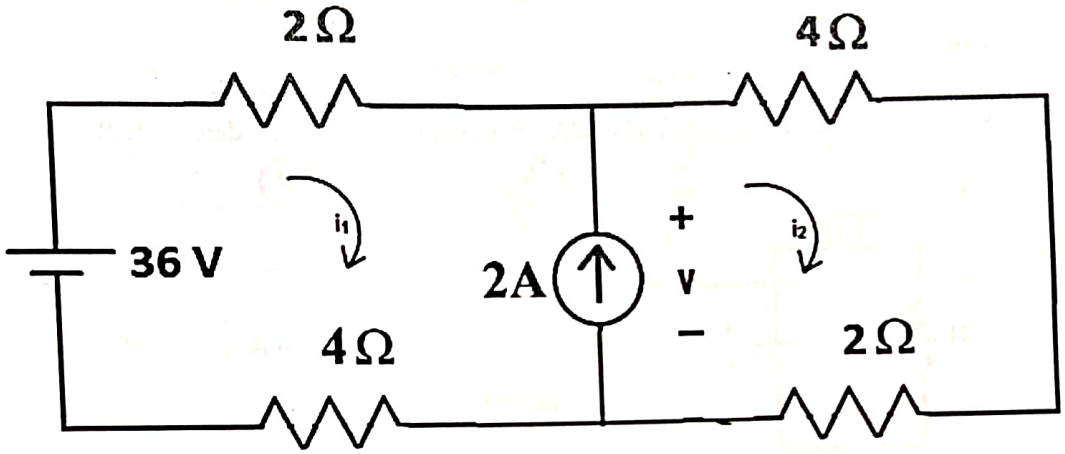
NETWORK ANALYSIS [ECE 2122]

Marks: 30 **Duration: 90 mins.**

MCQ

Answer all the questions. Section Duration: 20 mins

1) In the circuit shown, voltage V across 2A current source is ---



(1)

1) 24 V 2) 12 V 3) 36 V 4) 60 V

Correct option is: 1

2) The following results were obtained from measurements taken between the two terminals of a resistive network. (1)

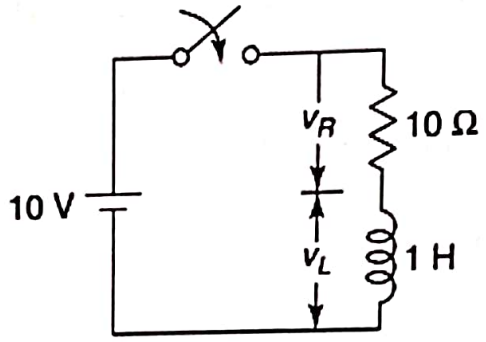
Terminal Voltage	12 V	0 V
Terminal Current	0 A	1.5 A

Thevenin equivalent of the network consists of

1) 12 V in series with 8 Ohm resistor
2) 1.5 A in series with 8 Ohm resistor
3) 12 V in parallel with 8 Ohm resistor
4) 8 V in series with 0 Ohm resistor.

Correct option is: 1

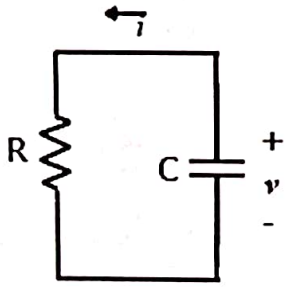
3) A series RL circuit shown in figure, 10V is applied at t = 0. At what time does $V_R = V_L$



1)	2.0693 S		2)	0.0693 S		3)	0 S		4)	1.0693 S	

Correct option is: 2

4) For the cicuit shown in figure if $v = 20e^{-2t}$ V and $i = 0.5\,e^{-2t}$ A, determine R and C values.



1) 40 Ohms, 12.5 F
2) 40 Ohms, 50 mF
3) 40 Ohms, 12.5 mF
4) 20 Ohms, 5 mF

Correct option is: 3

5) A 100 μF capacitor, carrying an initial charge of 500 μC, is discharging through a 50Ω resistor. Determine the time at which capacitor discharges to 185 μC.

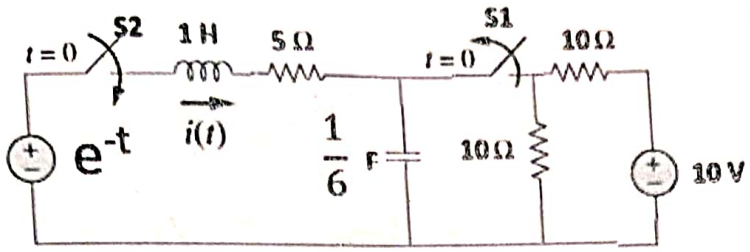
1) 2 mS 2) 0.5 mS 3) 10 mS 4) 5 mS

Correct option is: 4

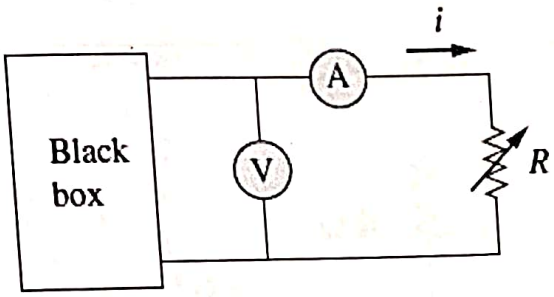
DESCRIPTIVE

Answer all the questions.

6) In the circuit shown below, at t = 0 switch S2 closed and switch S1 opened, before which a steady state has been reached. Find it) for t > 0. (4)



7) A black box with a circuit in it is connected to a variable resistor. An ideal ammeter (with zero resistance) and an ideal voltmeter (with infinite resistance) are used to measure current and voltage, as shown in Figure. The results are shown in the table.



$R(\Omega)$	$V(V)$	$i(A)$
2	3	1.5
8	8	1.0
14	10.5	0.75

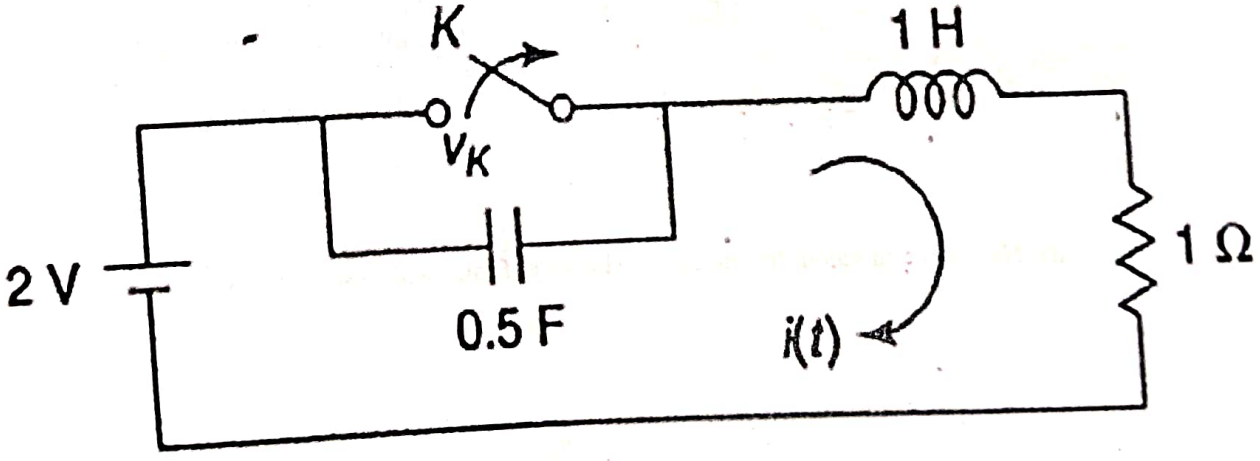
(3)

(a) Find i when R =4 ohm

(b) Determine the maximum power from the box.

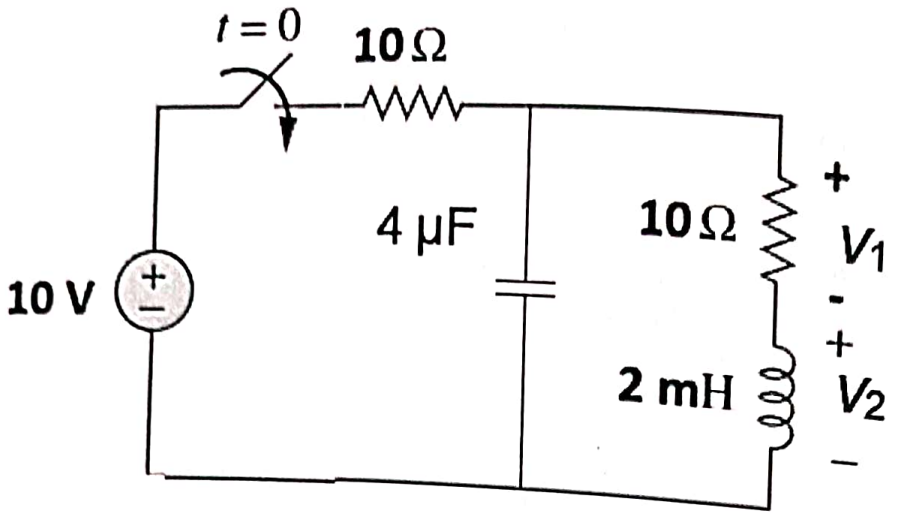
Use the maximum power transfer theorem.

8) The network shown attains steady state with the switch closed. At t = 0, the switch is opened. Find the voltage across the switch v_k and dv_k/dt at $t = 0^+$.

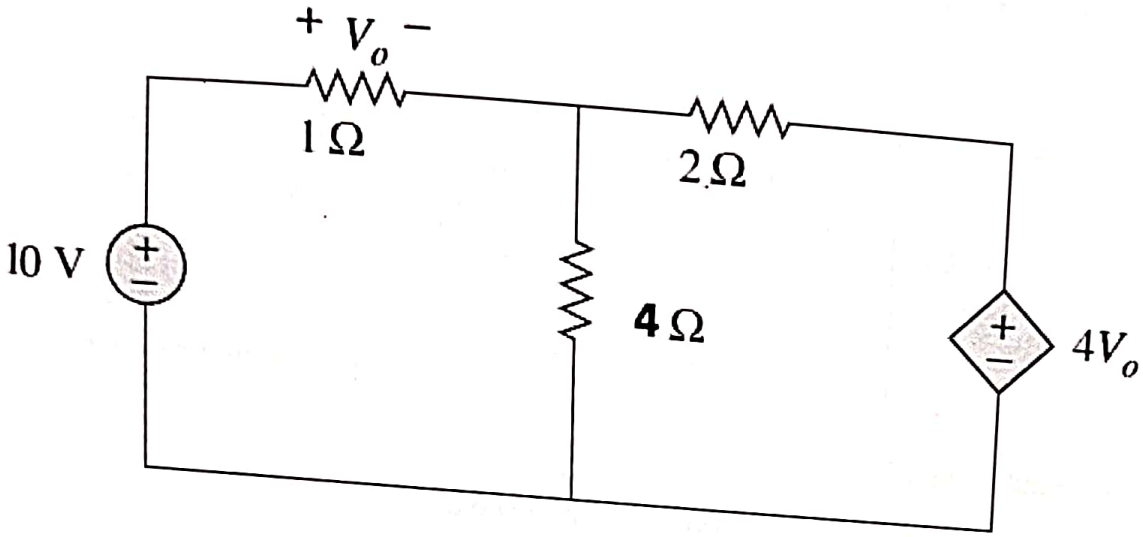


(3)

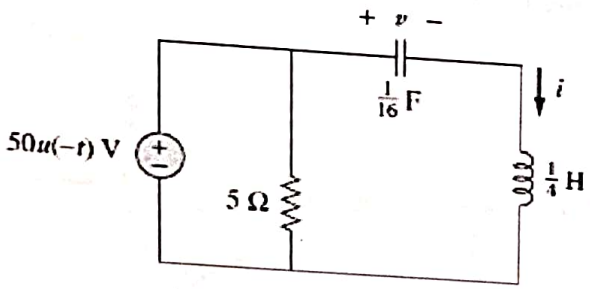
9) In the circuit shown, the switch is closed at t = 0. Find (a) $v_1(0^+)$, (b) $v_2(0^+)$, (c) $v_1(\infty)$ and (d) $v_2(\infty)$. (3)



10) Find current in 4 Ω resistor by applying Thevinins theorem.

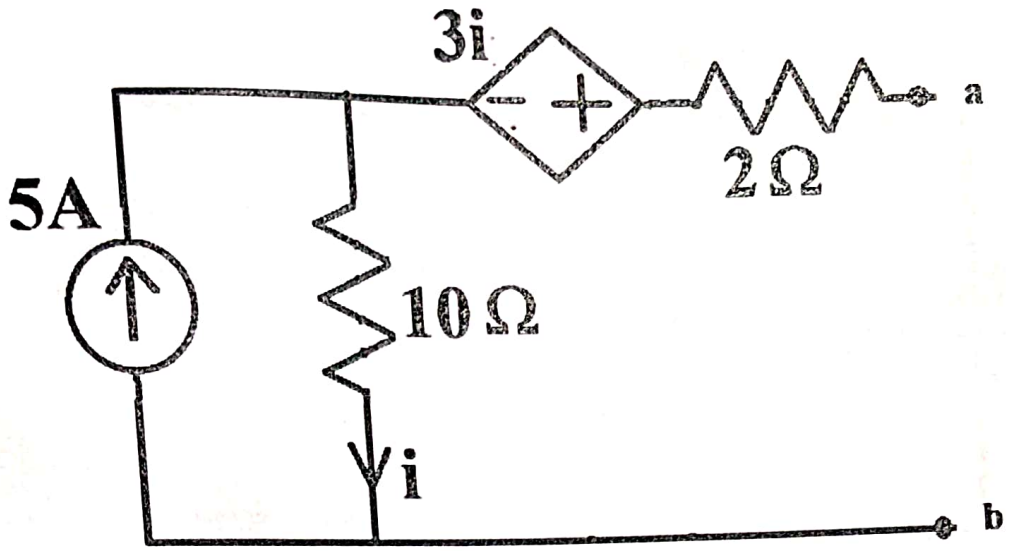


11) For the circuit shown , find i(t) for t ≥ 0.

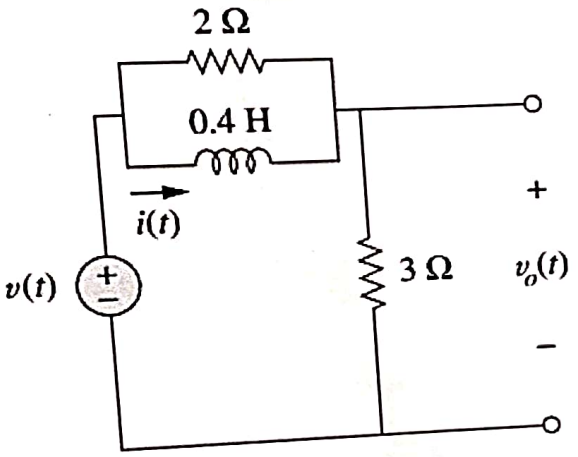


12) Find the Norton's equivalent for the circuit shown in figure with respect to ab.

(2)

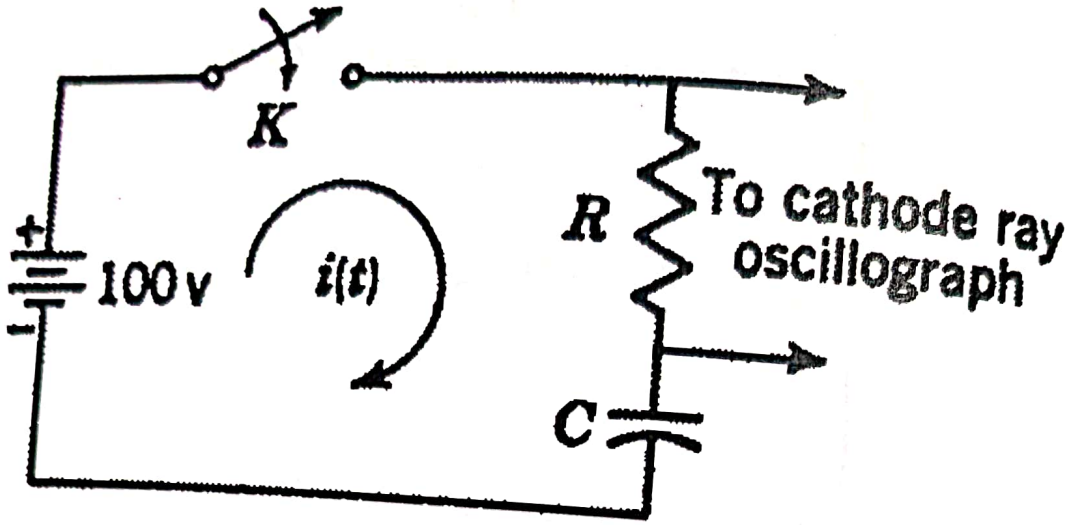


13) In the network shown in Figure, find $v_O(t)$, if i(0) =5 A and $v(t)$ = 0 V



(2)

14) In the circuit shown in the figure, the switch K is closed at t=0. The current waveform is observed with a cathode ray oscilloscope. The initial value of current is measured to be 0.01 amp. The transient appears to disappear in 0.1 sec. Find (a) the value of R, (b) the value of C, and (c)the equation of i(t). (2)



-----End-----

Scheme of valuation

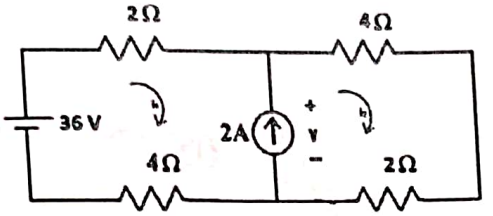
B.Tech III Semester Sessional Examination September 2024

NETWORK ANALYSIS [ECE 2122]

Exam Date & Time: 25-Sep-2024 (10:45 AM - 12:15 PM)

Marks: 30 **Duration: 90 mins.**

1) In the circuit shown, voltage V across 2A current source is --- (1)



1) 24 V 2) 12 V 3) 36 V 4) 60 V

Ans: 24 V

2) The following results were obtained from measurements taken between the two terminals of a resistive network. (1)

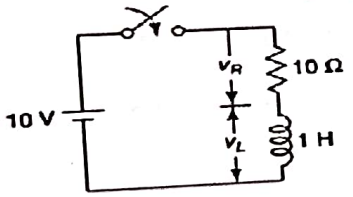
Terminal Voltage	12 V	0 V
Terminal Current	0 A	1.5 A

Ans: 12 V in series with 8 ohm resistor

Thevenin equivalent of the network consists of

1) 12 V in series with 8 Ohm resistor 2) 1.5 A in series with 8 Ohm resistor 3) 12 V in parallel with 8 Ohm resistor 4) 8 V in series with 0 Ohm resistor.

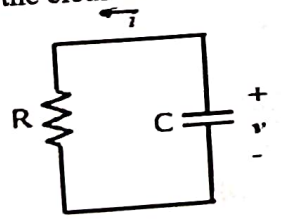
3) A series RL circuit shown in figure, 10V is applied at t = 0. At what time does $V_R = V_L$ (1)



1) 2.0693 S 2) 0.0693 S 3) 0 S 4) 1.0693 S

Ans: 0.0693 S

4) For the cicuit shown in figure if $v = 20e^{-2t}$ V and $i = 0.5\,e^{-2t}$ A, determine R and C values. (1)



1. 40 Ohms, 12.5 F
2. 40 Ohms, 50 mF
3. 40 Ohms, 12.5 mF
4. 20 Ohms, 5 mF

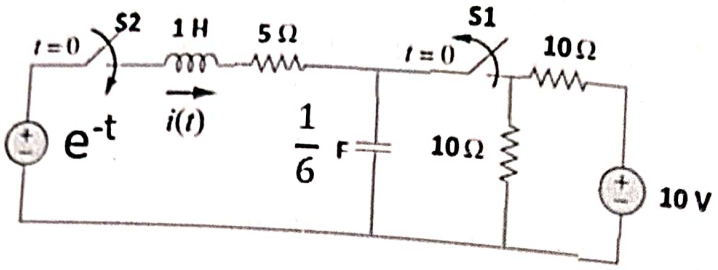
Ans: 40 Ohms, 12.5 mF

5) A 100 μF capacitor, carrying an initial charge of 500 μC, is discharging through a 50Ω resistor. (1)
Determine the time at which capacitor discharges to 185 μC.

1) 2 mS 2) 0.5 mS 3) 10 mS 4) 5 mS

Ans: 5 mS

6) In the circuit shown below, at t = 0 switch S2 closed and switch S1 opened, before which a steady state has been reached. Find it) for t > 0. (4)



Solution: At t = 0⁻ $v_c(0^-) = 5\,V$

At t = 0 + $i(0^+) = 0\,A\ ;\ v_c(0^+) = 5\,V$

For t ≥ 0, $\frac{di}{dt} + 5i + \frac{1}{1/6}\int i\,dt = e^{-t}$ (1)

$\frac{di}{dt} = -5i - 6\int i\,dt + e^{-t}$ (2)

$\frac{d\,i(0^+)}{dt} = -5\,i(0^+) - v_c(0^+) + e^{-0} = -4\,A/Sec$ (3) [1Mark]

Differentiating (1) $\frac{d^2i}{dt^2} + 5\frac{di}{dt} + 6i = -e^{-t}$ (4)

Characteristic equation $s^2 + 5s + 6 = 0, with\ roots\ s_1 = -2\ and\ s_2 = -3$

Complementary solution is $i_c(t) = k_1 e^{-2t} + k_2 e^{-3t}$ (5)

Particular solution is $i_p(t) = A e^{-t}$ (6) [1Mark]

Substitute (6) in (4) $A e^{-t} - 5A e^{-t} + 6A e^{-t} = -e^{-t} \rightarrow A = -\frac{1}{2}$

Therefore, $i_p(t) = -\frac{1}{2} e^{-t}$

[1 mark]

And

$i(t) = k_1 e^{-2t} + k_2 e^{-3t} - \frac{1}{2}e^{-t}$ (7)

$I(0) = 0 = k_1 e^0 + k_2 e^0 - \frac{1}{2}e^0$

$k_1 + k_2 = \frac{1}{2}$ (8)

Differentiate (7) $\frac{di(t)}{dt} = -2k_1 e^{-2t} - 3k_2 e^{-3t} - \frac{1}{2}e^{-t}$

$\frac{di(0)}{dt} = -4 = -2k_1 e^0 - 3k_2 e^0 - \frac{1}{2}e^0$

$2k_1 - 3k_2 = \frac{9}{2}$ (9)

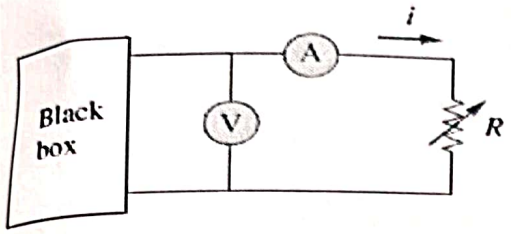
Solving (8) and (9) $k_1 = -3\ \ and\ k_2 = \frac{7}{2}$

Total solution

$i(t) = -3\,e^{-2t} - 3.5\,e^{-3t} - \frac{1}{2}e^{-t}\quad for\ t \geq 0$

[1 mark].

7) A black box with a circuit in it is connected to a variable resistor. An ideal ammeter (with zero resistance) and an ideal voltmeter (with infinite resistance) are used to measure current and voltage, as shown in Figure. The results are shown in the table. (3)

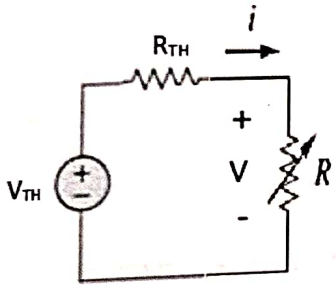


$R(\Omega)$	$V(V)$	$i(A)$
2	3	1.5
8	8	1.0
14	10.5	0.75

(a) Find i when R =4 ohm

(b) Determine the maximum power from the box.
Use the maximum power transfer theorem.

Solution:



$V_{TH} = V + iR_{TH}$ ----------(1 Mark)

Case1) When i = 1.5, V= 3 $V_{TH} = 3 + 1.5R_{TH}$ –(1)

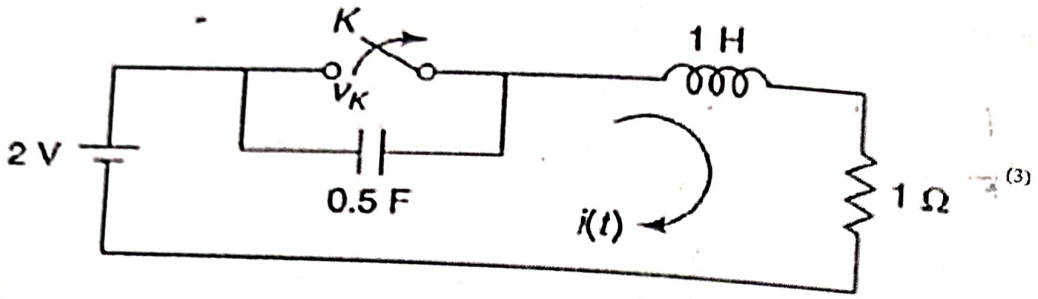
Case2) When i = 1, V= 8 $V_{TH} = 3 + 1.R_{TH}$ --(2)

Solving (1) and (2) we get $V_{TH} = 18\ V$ *and* $R_{TH} = 10\ \Omega$ ----------(1 Mark)

(a) R=4Ω $I = V_{TH}/4 + R_{TH} = 18/(4+10) = 1.285\ A$ ----------(0.5 Marks)

(b) *For maximum power transfer* $P = V_{TH}^2/4R_{TH} = 8.1\ W$ ----------(0.5 Marks)

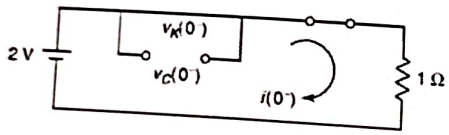
8) The network shown attains steady state with the switch closed. At t = 0, the switch is opened. Find the voltage across the switch v_K and dv_K/dt at $t = 0^+$.



(3)

Solution:

At $t = 0^-$, the network attains steady-state condition. The capacitor acts as an open circuit and the inductor acts as a short circuit.



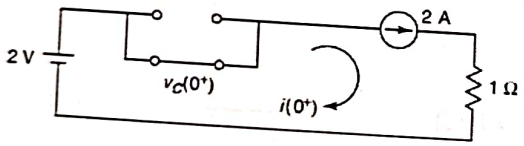
Equivalent Circuit at $t = 0^-$

$$i(0^-) = \frac{2}{1} = 2\ \text{A}$$

$$v_C(0^-) = 0$$

--------- (1 Mark)

At $t = 0^+$ (switch is opened), the capacitor acts as a short circuit and the inductor acts as a current source of 2 A.



Equivalent Circuit at $t = 0^+$

$$i(0^+) = 2\ \text{A}$$

$$v_C(0^+) = 0$$

$$v_K(0^+) = 0$$

$v(t) = \frac{1}{C}\int_0^t i(t)dt + v(0)$

$$v_K = \frac{1}{C}\int i\,dt$$

Differentiating ⟶

$$\frac{dv_K}{dt} = \frac{i}{C}$$

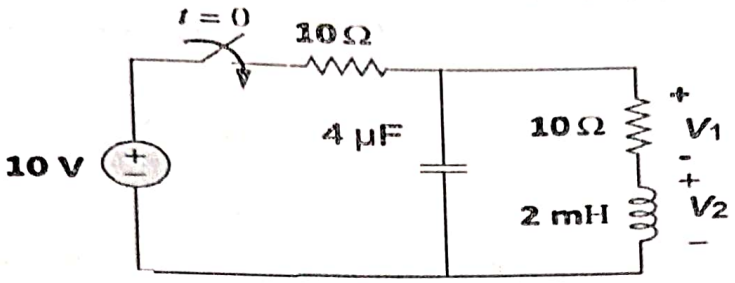
At $t = 0^+$ ⟶

$dv_K/dt(0+) = i(0+)/C$

$= 2/0.5 = 4V/s$

---------- (2 Marks)

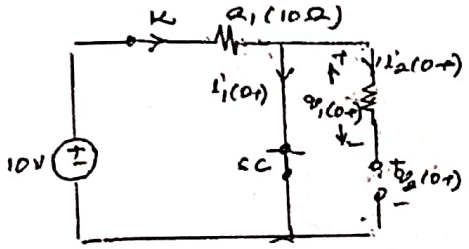
9) In the circuit shown, the switch is closed at t = 0. Find (a) $v_1(0^+)$, (b) $v_2(0^+)$, (c) $v_1(\infty)$ and (d) $v_2(\infty)$.



(3)

Solution: $v_C(0^-) = 0 = v_C(0^+)$; $i_L(0^-) = 0 = i_L(0^+)$

At t = 0⁺, Capacitor acts as Short Circuit and inductor acts as Open Circuit. --------- (1 Mark)

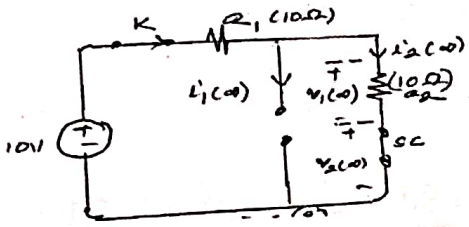


⟶ At t = 0⁺, Capacitor acts as Short Circuit and inductor acts as Open Circuit.

$i_1(0^+) = 10/R_1 = 1A$
$i_2(0^+) = 0$

$v_1(0^+) = 0V$
$v_2(0^+) = 0V$

--------- (1 Mark)



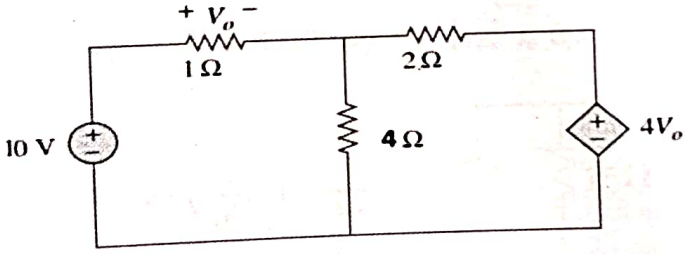
⟶ At t = ∞ , Capacitor acts as open Circuit and inductor acts as Short Circuit.

$i_2(\infty) = 10 / (R_1 + R_2) = 0.5A$
$i_1(\infty) = 0$

$v_1(\infty) = i_2(\infty) . R_2 = 5V$
$v_2(\infty) = 0V$

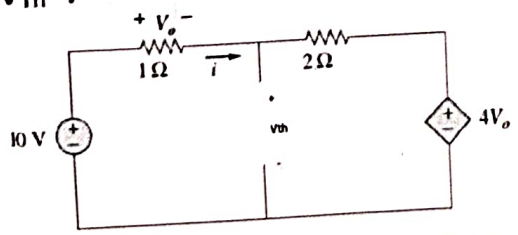
--------- (1 Mark)

10. Find current in 4 Ω resistor by applying Thevenin's theorem.



Solution:

V_{Th} :



$10 - i - 2i - 4Vo = 0;$

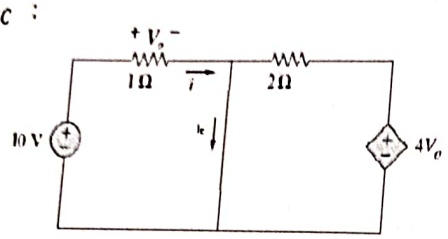
$Vo = i$

$i = \frac{10}{7} A$

$V_{TH} = 10 - 1i = \frac{60}{7} V$

(1 Marks)

I_{SC} :



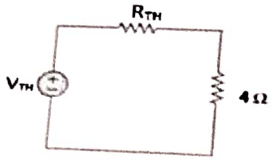
$$I_{sc} = \frac{10}{1} + \frac{4Vo}{2} \quad and\ Vo = i = \frac{10}{1} = 10A$$

$$I_{sc} = \frac{10}{1} + \frac{40}{2} = 30A$$

(1 Marks)

$$R_{TH} = \frac{V_{TH}}{I_{SC}} = \frac{2}{7}\ Ohms$$

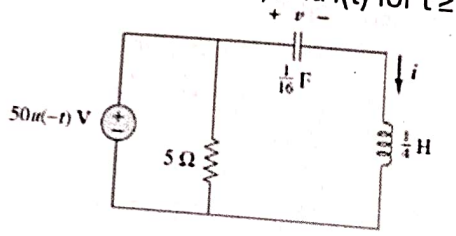
(0.5 Marks)



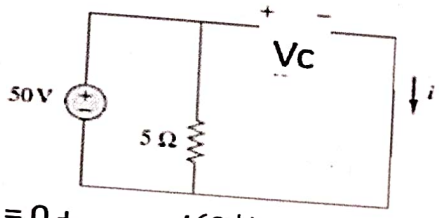
$$I_{4\ Ohms} = \frac{60/7}{2/7 + 4} = 2\ A$$

(0.5 Marks)

11. For the circuit shown , find i(t) for t ≥ 0



At t = 0^-

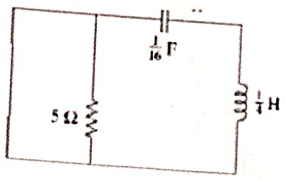


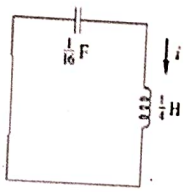
$$v_c(0^-) = 50\ V \ \ and \ \ i(0^-) = 0A$$

At t = 0 +

$$i(0^+) = 0\ A\ ;\ v_c(0^+) = 50\ V$$

For t ≥ 0





$$\frac{1}{4}\frac{di}{dt} + \frac{1}{1/16}\int i\,dt = 0 \qquad (1)$$

$$\frac{di}{dt} = 4(0 - 16\int i\,dt)$$

$$\frac{d\,i(0^+)}{dt} = 4(0 - v_c(0^+)) = -200\ A/Sec$$

---------- [1Mark]

Differentiating (1) $\frac{d^2i}{dt^2} + 64i = 0$ (2)

Characteristic equation $s^2 + 64 = 0, with\ roots\ s_{1,2} = \pm j8$

$i(t) = k_1 cos8t + k_2\ sin8t$ (3) ---------- [1Mark]

$i(0) = 0 = k_1\ cos0 + k_2\ sin0$

$k_1 = 0$

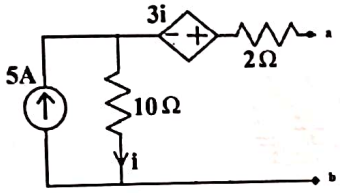
Differentiate (3) $\frac{di(t)}{dt} = -8k_1\ sin8t + 8k_2 cos8t$

$\frac{di(0)}{dt} = -200 = -8k_1 sin0 + 8k_2\ cos0$

$k_2 = -25$

Therefore $i(t) = -25cos8t$ ---------- [1Mark]

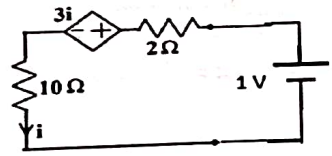
12. Find the Norton's equivalent for the circuit shown in figure with respect to ab.



Solution:

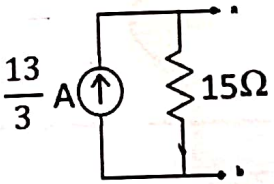
V_{OC} : From the network $i = 5A$; $V_{OC} = 10i + 3i = 65\ V$ ----------(0.5 marks)

R_{TH} :



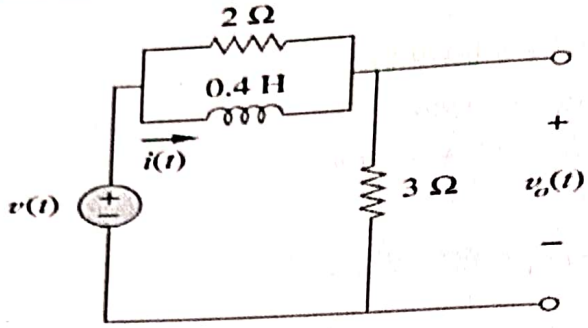
$3i + 2i + 10i = 1;\quad i = \frac{1}{15}\ A \qquad R_{TH} = \frac{1}{i} = 15\ Ohms$ ----------(1 Mark)

$I_{SC} = \frac{V_{OC}}{R_{TH}} = \frac{65}{15} = \frac{13}{3}\ A$

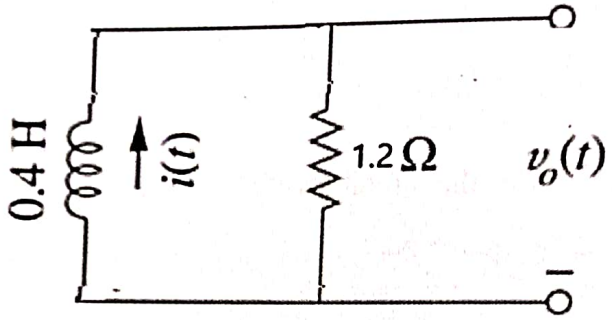


----------(0.5 marks)

13) In the network shown in Figure, find $v_O(t)$, if i(0) =5 A and $v(t) = 0$ V (2)



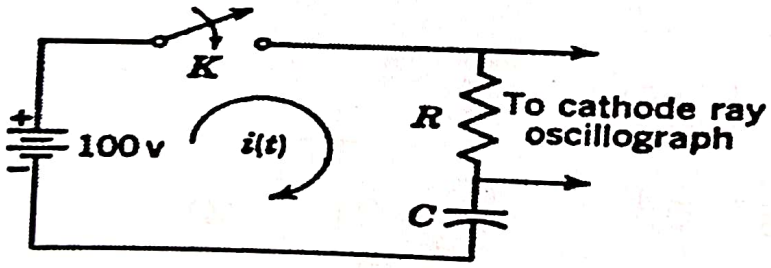
Solution: $v(t) = 0$ *and* $i(0) = I_o = 5A$



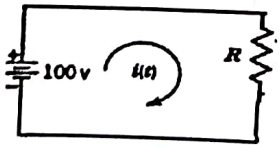
$R_{eq} = 2||3\ \ 1.2\ Ohms$; $\quad i(t) = I_o e^{\frac{-Rt}{L}} = 5\,e^{-3t} - - - - - (1\ mark)$

$V_O(t) = -L\frac{di}{dt} = -(0.4)(-3)(5)e^{-3t} = 6\,e^{-3t} = Volts - - - - - - (1\ mark)$

14) In the circuit shown in the figure, the switch K is closed at t=0. The current waveform is observed with a cathode ray oscilloscope. The initial value of current is measured to be 0.01 amp. The transient appears to disappear in 0.1 sec. Find (a) the value of R, (b) the value of C, and (c)the equation of i(t). (2)



Solution: $At\ t = 0^+$;



(a) $i(0) = 0.01 = \frac{100}{R}$; $\quad R = 10\ kOhms$ ----(0.5 marks)

(b)*Transient disappears in 4 Time constants, ie* $4\tau = 4RC = 0.1 \rightarrow\rightarrow C = 2.5\mu F$ - ---- (0.5 marks)

(c) $i(t) = i(0)e^{\frac{-t}{RC}} = 0.01\,e^{-40t}$ *Amps* ---------- (1 mark)